

48.3

$$Z = 5000X_1 + 4000X_2 + 2000X_3 \quad Z = 5000X_1 + 4000X_2 + 2000X_3 + 0X_4 + 0X_5 + 0X_6 + 1X_7 + 1X_8 + 1X_9$$

$$2X_2 + X_3 \geq 20 \quad 2X_2 + X_3 - X_4 + X_7 = 20$$

$$2X_1 + 4X_2 \geq 8 \quad 2X_1 + 4X_2 - X_5 + X_8 = 8$$

$$5X_1 + 2X_2 + X_3 \geq 30 \quad 5X_1 + 2X_2 + X_3 - X_6 + X_9 = 30$$

B	CB	P	X_1 5000	X_2 4000	X_3 2000	X_4 0	X_5 0	X_6 0	X_7 1	X_8 1	X_9 1
X_7	M	20	0	2	1	-1	0	0	1	0	0
X_8	M	8	2	4	0	0	-1	0	0	1	0
X_9	M	30	5	2	1	0	0	-1	0	0	1
min	58M		7M-5000	8M-4000	2M-2000	-M	-M	-M	0	0	0
X_7	M	16	-1	0	1	-1	1/2	0	1	-1/2	0
X_2	4000	2	1/2	1	0	0	-1/4	0	0	1/4	0
X_9	M	26	4	0	1	0	1/2	-1	0	-1/2	1
min	18000 42M+2000		3M-3000	0	2M-2000	-M	M-1000	-M	0	2M+1000	0
X_7	M	20	0	2	1	-1	0	0	1	//	0
X_1	5000	4	1	2	0	0	-1/2	0	0	//	0
X_9	M	10	0	-8	1	0	5/2	-1	0	//	1
min	18000 30M		0	6M+6000	2M-2000	-M	5M-2000	-M	0	//	0
X_7	M	20	0	2	1	-1	0	0	1	//	0
X_1	5000	6	1	2/5	1/5	0	0	-1/5	0	//	1/5
X_5	0	4	0	-16/5	2/5	0	1	-2/5	0	//	2/5
min	18000 20M		0	2M-2000	M-1000	-M	0	-1000	0	//	1000+M
X_2	4000	10	0	1	1/2	-1/2	0	0	1/2	//	//
X_1	5000	2	1	0	0	-1/5	0	-1/5	-1/5	//	//
X_5	0	36	0	0	2	-8/5	1	-2/5	//	//	//
min	50000		0	0	0	-1000	0	-1000	//	//	//

$$Z = 50000, X_1 = 2, X_2 = 10, X_3 = 0, X_4 = 0, X_5 = 36$$

$$Z = 5000 \cdot 2 + 4000 \cdot 10 + 2000 \cdot 0 = 50000 \checkmark$$